

Huaying Yuan

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Education

- **Renmin University of China**; PhD candidate in Artificial Intelligence; 2022.09-2027.06 (Expected)
 - **Supervisor**: Prof. Zhicheng Dou and Prof. Ji-Rong Wen
 - **GPA**: 3.87/4.0; Intelligent Information Retrieval(A); Machine Learning(A); Artificial Intelligence(A)
- **Nankai University**; BS in Computer Science 2018.09-2022.06
 - **GPA**: 91.64/100, **RANK**: 2/116; Calculus(96); Computer Programming(94); Data Structure(95)
 - **CET 6**: 538; **CET 4**: 585

Research Interest

My research interests focus on **video understanding**, **multi-modal understanding** and **video retrieval**.

Publications

- **Memory-enhanced Retrieval Augmentation for Long Video Understanding** [Paper](#) In submission
H Yuan, Z Liu, M Qin, Hongjin Qian, Y Shu, Z Dou, JR Wen
Motivation: Retrieval-augmented generation (RAG) shows promise for long-video understanding (LVU) but struggles without explicit queries. To tackle this, we introduce MemVid, a novel RAG-based LVU approach inspired by human cognitive memory, which follows a memorizing-reasoning-retrieving-focusing process.
- **MomentSeeker: A Comprehensive Benchmark and A Strong Baseline For Moment Retrieval Within Long Videos** [Paper](#) In submission
H Yuan, J Ni, Y Wang, J Zhou, Z Liang, Z Liu, Z Cao, Z Dou, JR Wen
Motivation: Retrieval-augmented generation aids long-video understanding but lacks a benchmark. We introduce MomentSeeker, a benchmark for long-video moment retrieval with 500s+ videos, diverse tasks, and human evaluation, along with an MLLM retriever fine-tuned on synthetic data for strong performance.
- **Omnigen: Unified image generation** [Paper](#) CVPR 2025
S Xiao, Y Wang, J Zhou, H Yuan, X Xing, R Yan, S Wang, T Huang, Z Liu
Motivation: Large Language Models have unified language generation, but a unified model for image generation remains unexplored. In this paper, we introduce OmniGen, a unified diffusion model for image generation that supports text-to-image, image editing, and visual-conditional generation. Its simplified architecture enables end-to-end generation without plugins.
- **FineRAG: Fine-grained Retrieval-Augmented Text-to-Image Generation** [Paper](#) COLING 2024
H Yuan, Z Zhao, S Wang, S Xiao, M Ni, Z Liu, Z Dou
Motivation: Recent text-to-image models like Stable Diffusion produce diverse, high-quality images but still lack robust knowledge access. To tackle this problem, we introduce FineRAG, a fine-grained model that systematically breaks down the retrieval-augmented image generation task into four critical stages: query decomposition, candidate selection, retrieval-augmented diffusion, and self-reflection.
- **VILE: Block-Aware Visual Enhanced Document Retrieval** [Paper](#) CIKM 2023
H Yuan, Z Dou, Y Zhou, Y Guo, JR Wen
Motivation: Document retrieval is crucial in web search, but recent models focus only on textual content, ignoring the visual aspects of web pages. We propose VILE, a dense retrieval model that incorporates visual features into document representations.

Competition

- The 46th International Collegiate Programming Contest (ICPC) Asia Shenyang Regional Contest; **Silver Medal**
- The 46th International Collegiate Programming Contest (ICPC) Asia Kunming Regional Contest; **Bronze Medal**
- National College Students Mathematical Contest in Modeling; **National second prize**

Awards

- First-class Scholarship; Gaoling School of Artificial Intelligence; 2022.12
- Renmin University of China First-class Scholarship (5%); Nankai University; 2021.12, 2020.12
- National Scholarship (1.2%); Nankai University; 2019.12